

Reliable LNG Fuel Supply through Advanced LNG Pump Technology



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01

About PANASIA EM

1. About PANASIA EM

PANASIA Group

PANASIA Headquarters



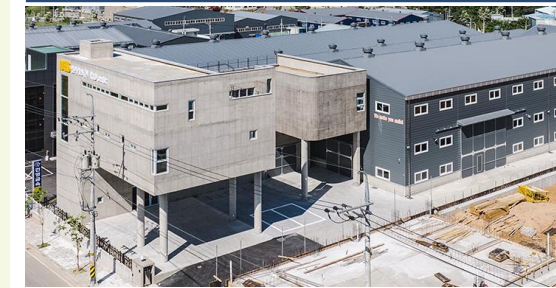
A Leading Marine Environmental Equipment Provider
55, Mieumsandan 3ro, Gangseo-gu, Busan, Republic of Korea

PANASIA EM



Specialized Manufacturer of LNG DF Products

NES Inc. (Subsidiary of PANASIA EM)



Expansion Joint & Dampers Manufacturer

Shanghai (China)



Dalian (China)



Osaka (Japan)



Rotterdam (Netherlands)



PANASIA Sales & After-sales Service Station

+ To be opened at
Athens (Greece), Singapore (Singapore), Rugao (China)

1. About PANASIA EM

PANASIA Product

WATER Solutions



Measurement &
Control System



Ballast Water
Treatment System



Water Treatment
System

AIR Solutions



De-SOx
Scrubber



De-NOx
SCR System



Engine Exhaust Recycling
System (iCER)

ENERGY Solutions



AMP
Alternative Maritime
Power System



Carbon Capture and
Storage System



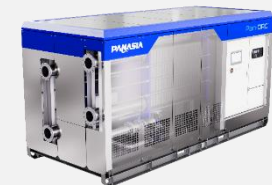
CEMS



N₂GEN



Fuel Supply System
(LNG/Methanol/Ammonia)



ORC

1. About PANASIA EM

PANASIAEM Product

LNG Fuel Pump(LP/HP)



LNG BOG Compressor



CO2 Liquefaction for OCCS





02

Global LNG Dual-Fuel Market

2. Global LNG Dual-Fuel Market

IMO 2020 Environmental Regulation Overview

International Maritime Organization(IMO) strictly limiting Sulfur Oxides(SOx) emissions from marine fuels(MARPOL Annex VI)

Fuel Sulfur Limit : 3.5% → 0.5% or less

Emission Control Areas(ECA) : 0.1% or less

Solutions for IMO 2020

1. Very Low Sulfur Fuel Oil (VLSFO)

No retrofitting cost required, but increases OPEX due to higher fuel costs.
Incapable of meeting long-term Greenhouse Gas (GHG) reduction targets.

2. Scrubber Installation

Existing high-sulfur fuel oil can be used, but this will result in initial retrofitting costs and a loss of cargo space.

3. Adoption of LNG Fuel

A practical and immediate solution to meet IMO 2020 regulations using proven marine fuel technology.

Alternative Fuel Comparison

Fuel Type	Maturity Level	Infrastructure	Strengths	Limitations
LNG	●	●	Supply Chain Maturity SOx Reduction	Methane Slip Cryogenic Storage
Methanol	▲	▲	Easy Storage & Handling	Low Energy Density High Eco-friendly Fuel Price
Ammonia	X	▲	No CO ₂ during combustion	Toxicity, Ignitability, Safety Issues
Hydrogen	X	X	Carbon-free Fuel	Storage Difficulty Low Volumetric Energy Density

✓ Conclusion : LNG remains a practical and mature transition fuel.

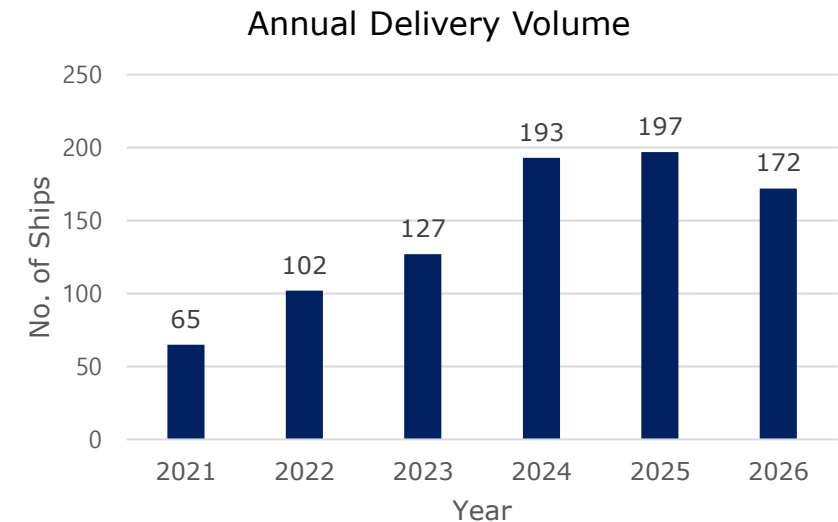
2. Global LNG Dual-Fuel Market

Global LNG DF Vessel Delivery(2021~2026)

(Excluding LNGC)

Unit : Ships

Ship Type	2021	2022	2023	2024	2025	2026	Total	Share
Container	14	14	28	74	85	44	259	30.3%
Car carrier	11	18	12	51	65	47	204	23.8%
Tanker	21	33	37	21	16	47	175	20.5%
Bulker	1	7	26	16	16	15	81	9.5%
ETC	8	19	8	15	5	6	61	7.1%
Cruise Ship	4	5	6	5	7	6	33	3.9%
LNG Bunkering	4	4	6	6	3	6	29	3.4%
Tug	2	2	4	5	0	1	14	1.6%
Annual Total	65	102	127	193	197	172	856	100.0%



Total Cumulative Delivery (2021~2026)

856 Ships

For LNG DF Vessels(Excluding LNGC)

Rapid Increase in Annual Delivery

65 → 197 Ships

approximately 3 times in 2025 compared to 2021

Largest Share Ship Type

Container 259 Ships

Market Share abt. 30% in LNG DF Vessels

2. Global LNG Dual-Fuel Market

Global LNG DF Vessel Delivery(2027~2029)

(Excluding LNGC)

Unit : Ships

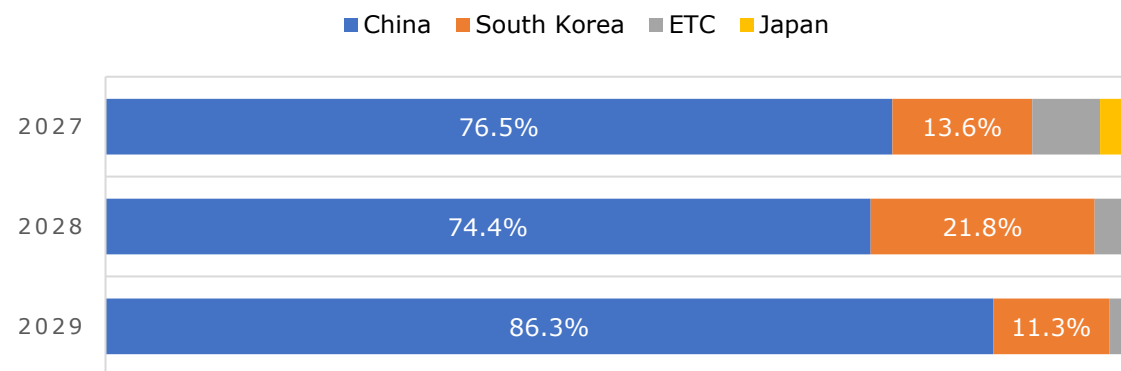
Ship Type	2027	2028	2029	Total	Share
Container	91	170	173	434	63.2%
PCTC/RORO	42	41	25	108	15.7%
Tanker	31	18	6	55	8%
LNG Bunkering	26	16	0	42	6.1%
ETC	4	8	5	17	2.5%
Cruise Ship	8	5	3	16	2.3%
Bulk Carrier	11	4		15	2.2%
Annual Total	213	262	212	687	100.0%

Korea/China/Japan Expected Delivery Comparison

(Excluding LNGC)

Unit : Ships

Country	2027	2028	2029	Total	Share
China	163	195	183	541	78.7%
South Korea	29	57	24	110	16%
Japan	7	3	0	10	1.5%
ETC	14	7	5	26	3.8%
Annual Total	213	262	212	687	100.0%



Total Cumulative Expected Delivery (2027~2029)

687 Ships

For LNG DF Vessels(Excluding LNGC)

Largest Share Ship Type

Container 434 Ships

Market Share abt. 63% in LNG DF Vessels

Market Share by Country(2027~2029)

China 78.7% / Korea 16%

China 541ships / Korea 110ships / Japan 10ships

An aerial photograph showing a dense, vibrant green forest on the left, which meets a calm, turquoise body of water on the right. The forest is composed of many small, rounded tree canopies. The water has a fine, rippled texture. A small, simple wooden pier or dock is visible at the edge of the forest where it meets the water.

03

Pros & Cons of LNG as a Marine Fuel

3. Pros & Cons of LNG as a Marine Fuel

Pros of LNG **LNG as a Proven Solution for Air Emission Compliance**



LNG significantly reduces local air pollutants and provides an effective solution for compliance with IMO air-emission regulations.

Key Benefits

- SO_x: Nearly eliminated due to negligible sulfur content
- Particulate Matter (PM): Nearly eliminated
- NO_x: Up to ~80% reduction depending on engine technology
- Particularly attractive for vessels operating in Emission Control Areas (ECAs)

Relevant Regulations

MARPOL Annex VI, Regulation 14 – SO_x

- Global sulfur limit: 0.50% m/m
- ECA sulfur limit: 0.10% m/m

MARPOL Annex VI, Regulation 13 – NO_x

Tier III requirements apply to applicable ships operating within designated NO_x ECAs

LNG offers a straightforward pathway to comply with increasingly stringent SO_x, PM and NO_x requirements

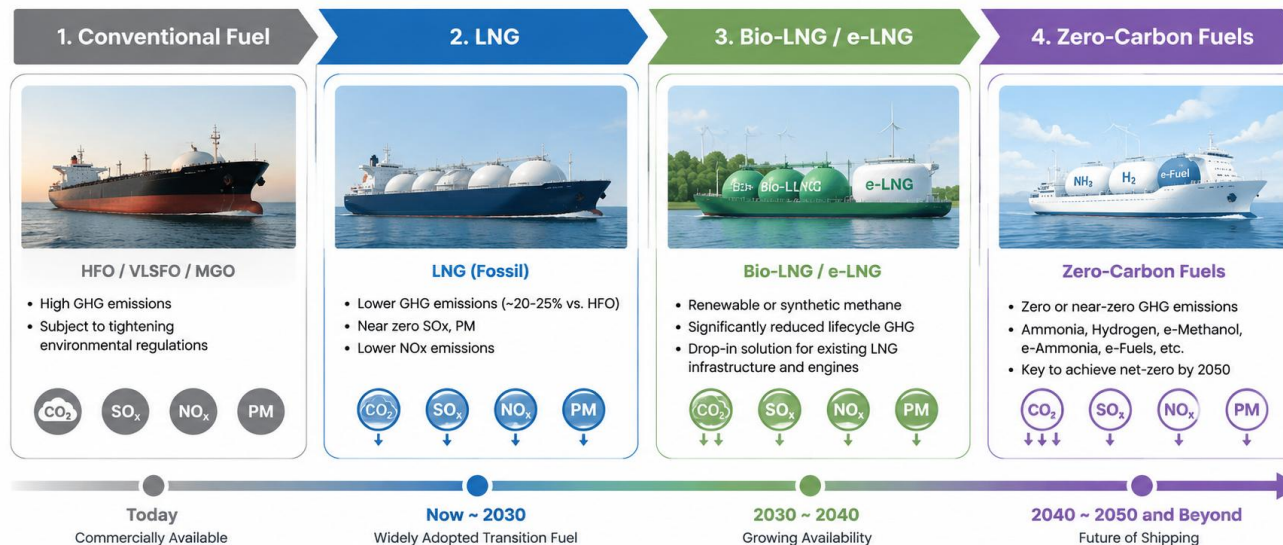
3. Pros & Cons of LNG as a Marine Fuel

Pros of LNG LNG as a Practical Transition Fuel

LNG combines near-term GHG reduction potential with mature technology and infrastructure, making it a practical transition fuel for shipping.

MARINE FUEL TRANSITION PATHWAY

Towards 2050 Net-Zero Shipping



Source : ICCT

Key Benefits

- Lower CO₂ emissions than conventional oil-based marine fuels due to LNG's lower carbon content
- Modern LNG propulsion can provide meaningful GHG reduction, depending on engine technology and methane slip
- Commercially proven dual-fuel engines and established global LNG supply infrastructure Existing LNG infrastructure can potentially support lower-carbon alternatives such as : Bio-LNG, e-LNG / synthetic methane

Transition Pathway

Fossil LNG(Today) -> Bio-LNG(Transition) -> e-LNG(Long-term)

Actual GHG benefits depend strongly on engine type, methane slip and the fuel's lifecycle emissions.

3. Pros & Cons of LNG as a Marine Fuel

Cons Of LNG

Methane Slip — The Key Challenge for LNG

Lower CO₂ emissions do not necessarily translate into lower overall GHG emissions.

Why Methane Matters

- LNG engines may release a portion of unburned methane during combustion
- Methane has a substantially higher climate impact than CO₂
- Methane slip can therefore offset part of LNG's CO₂ advantage
- The impact varies significantly by **engine technology and operating condition**

Engine Type	Default Methane Slip
Otto-cycle, DF medium-speed	3.1%
Lean-burn spark-ignited	2.6%
Otto-cycle, DF slow-speed	1.7%
Diesel-cycle, DF slow-speed	0.2%

(Source: FuelEU Maritime Regulation (EU) 2023/1805, Annex II)



Methane Slip captured by a Thermal Imaging Camera
(Source: European Federation for Transport and Environment)

The climate performance of LNG is highly dependent on engine technology and methane slip control.

3. Pros & Cons of LNG as a Marine Fuel

Cons Of LNG

Lifecycle Assessment Changes the LNG Equation

Marine fuel regulations are shifting from combustion-only CO₂ toward full lifecycle GHG emissions.

Traditional Approach	Emerging Regulatory Approach
Tank to Wake	Well to Wake
Fuel->Engine->Exhaust	Production->Processing->Transportation->Bunkering->Combustion
Main focus : CO ₂ from combustion	GHGs covered : CO ₂ +CH ₄ +N ₂ O

Relevant IMO Framework

- IMO Lifecycle GHG Assessment (LCA) Guidelines

- Lifecycle assessment consists of:

- Well-to-Tank (WtT)
 - Tank-to-Wake (TtW)

- Upstream methane leakage and onboard methane slip therefore become increasingly important.

Once lifecycle emissions are considered, LNG's GHG advantage can be significantly reduced.

Cons Of LNG

Tightening GHG Regulations Limit the Long-term Role of Fossil LNG

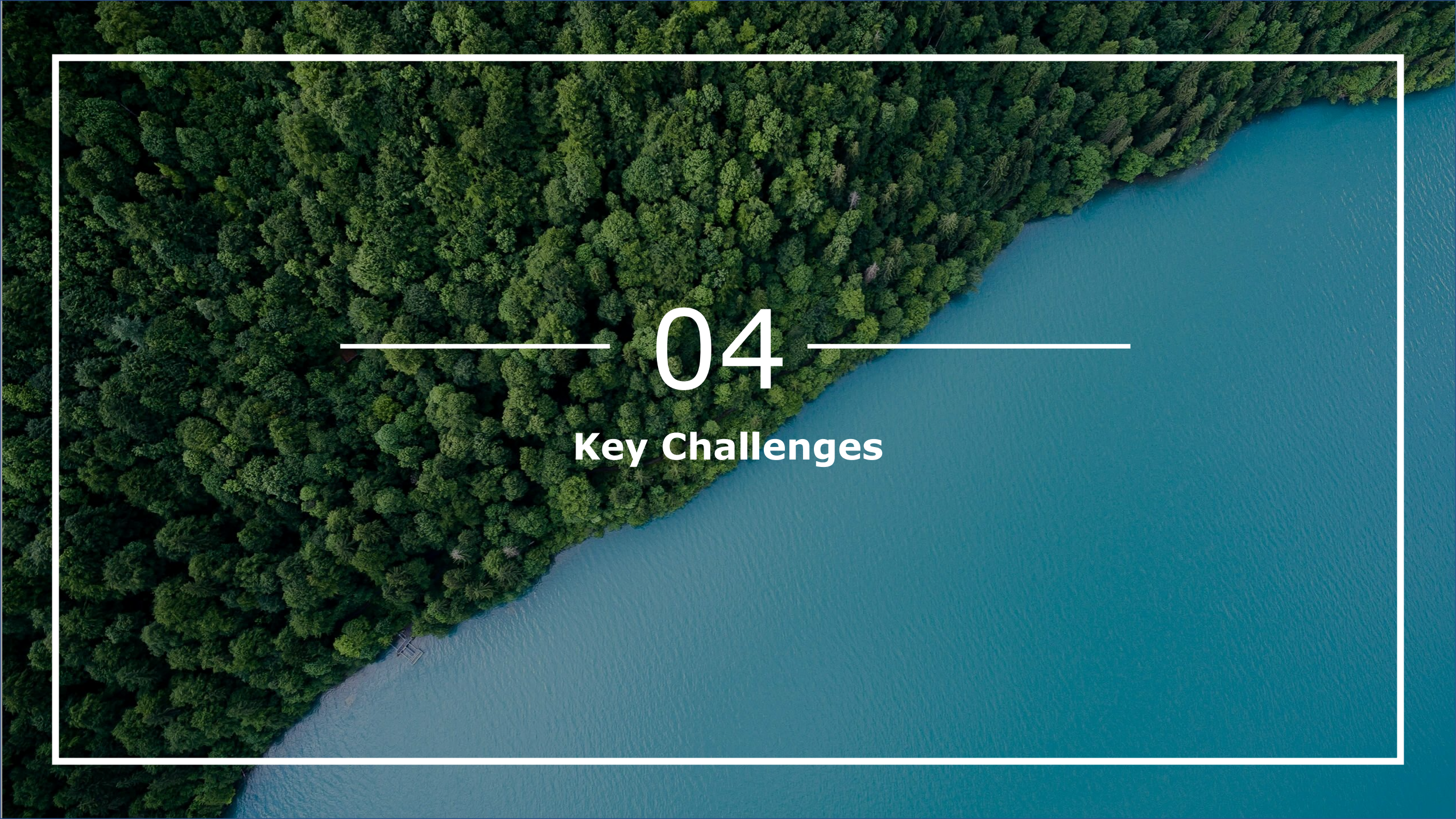
LNG can support near-term compliance, but tightening Well-to-Wake GHG requirements will increasingly challenge conventional fossil LNG.

FuelEU Maritime – GHG Intensity Reduction

Reference value: **91.16 gCO₂eq/MJ**

2025	2030	2035	2040	2045	2050
-2%	-6%	-14.5%	-31%	-62%	-80%

- Applied on a **Well-to-Wake GHG intensity** basis
- Includes **CO₂, CH₄ and N₂O**
- Methane slip is therefore directly reflected in LNG compliance performance



04

Key Challenges

5. Key Challenge

380 bar: A New Design Challenge for LNG Equipment

Everllence's next-generation ME-GI platform raises gas injection pressure from 300 bar to 380 bar, requiring the LNG fuel supply ecosystem to adapt accordingly. (Source : Everllence)

Market Update Note



5 March 2025

Launching ME-GI engines on 10.7 platform

**Introduction of MAN B&W
G95ME-C10.7-GI, G80ME-C10.7-GI,
G70ME-C10.7-GI, S60ME-C10.7-GI,
S50ME-C10.7-GI, and G50ME-C10.7-GI
engines**

The MAN B&W ME-GI engine is the leading engine for using methane as marine fuel for container vessels, product, chemical and crude tankers, bulk carriers, and PCTOs of all sizes. The superior fuel efficiency of the Diesel combustion cycle, combined with the lowest methane slip on the market, has established the ME-GI as the de facto standard methane-fuelled engine across these vessel segments.

Since the first ME-GI engines were delivered in 2014, hundreds have entered operation, collectively accumulating millions of running hours on gas.

The continuous development of the ME-GI engine resulted in the launch of the ME-GI Mk. 2 platform in 2020, which has become the standard design of today's shipbuilders. As of January 2025, the orderbook exceeds 850 ME-GI engines.

With the introduction of ME-GI engines on the 10.7 platform, we now present these as the latest generation of MAN B&W two-stroke engine designs.

The 10.7-GI engines come with:

- Increased supply and injection pressures.
- Higher fuel efficiency compared to previous GI variants.
- Engine designs based on the proven ME-GI Mk. 2 platform.

To further enhance fuel efficiency, the injection pressure has been increased from 300 bar on existing ME-GI engines to 380 bar on 10.7-GI engines.

To meet the new supply pressure, the MAN PrimeServ PVU (pump vaporiser unit) portfolio will be updated.

Tier III NO_x compliance
The new 10.7-GI engines enable a multitude of optional Tier III NO_x abatement solutions:

- G95ME-C10.7-GI - EcoEGR, EGRT[®], LPSCR[®]
- G80ME-C10.7-GI - EcoEGR, EGRT[®], HPSCR[®], LPSCR[®]
- G70ME-C10.7-GI - EcoEGR, HPSCR[®]
- S60ME-C10.7-GI - EcoEGR, HPSCR[®]
- S50ME-C10.7-GI - EcoEGR, HPSCR[®]
- G50ME-C10.7-GI - EcoEGR, HPSCR[®]

*includes sequential turbocharger cut-out (SEQ)

Fig. 1 on the next page shows layout diagrams of 10.7-GI engines currently available in the MAN B&W engine programme. Specific performance data and the exact availability of Tier III NO_x abatement technologies for the new engines are accessible in CEAS.

These new engines are available now, with the exact drawing timelines being evaluated at the time of ordering. The preliminary design drawing schedule is mid-2026 for the new 10.7-GI variants.

Questions regarding this Market Update Note should be directed to Two-stroke Promotion at Rasmus.Bildstrup@man-es.com.

MAN Energy Solutions
Tagholmegade 41
2450 Copenhagen SV, Denmark
www.man-es.com/marine

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- Increased supply and injection pressures.
- Higher fuel efficiency compared to previous GI variants.
- Engine designs based on the proven ME-GI Mk. 2 platform.

To further enhance fuel efficiency, the injection pressure has been increased from 300 bar on existing ME-GI engines to 380 bar on 10.7-GI engines.

380 bar: A New Design Challenge for LNG Equipment

3 Project Guide

3.1 Pump Vaporiser Unit

The Pump Vaporiser Unit (PVU) is a gas supply unit, which increases the gas pressure to the required supply pressure of the ME-GI / ME-GIE engine ranging from 250 bar to 380 bar and heats the second fuel (LFF) to vaporise to the gaseous form at 35 °C to 55 °C. The PVU is designed to supply LNG/ ethane to the pressure and temperature required by the Everllence B&W two-stroke ME-GI/GIE engines.

If the PVU used for any other purpose than stated above it is considered as misuse.

The PVU includes pumps (hydraulic actuator and cold-end pump), hydraulic accumulators, vaporiser and a gas strainer. Besides the unit is equipped with control cabinet, main operating panel and a control system with safety functions. The PVU is designed with a high-pressure reciprocating pump, consisting of three cylinders actuated by linear hydraulic pistons. The gas is in liquid form from the gas storage tank, through the cold end of the pump until it reaches the vaporiser. The PVU evaporates the liquid gas in the vaporiser and thus heats up the gas from approximately -160°C to 50°C for methane and from -90°C to 50°C for ethane. The vaporiser is a printed circuit heat exchanger.

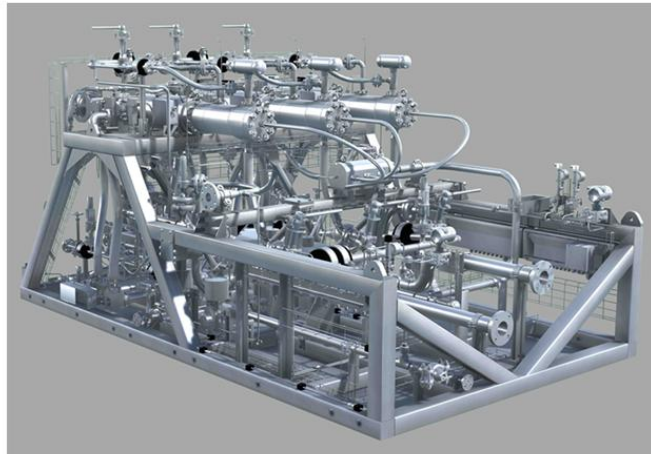


Figure 1: Pump Vaporiser Unit

3.1 Pump Vaporiser Unit

Project Guide

Why Higher Pressure?

- **Improved fuel injection / combustion performance**
- **Higher fuel efficiency**
- **Further optimization of high-pressure diesel-cycle LNG combustion**
- **Maintains the advantage of very low methane slip inherent to ME-GI architecture**

Challenge for Equipment Suppliers

The pressure increase does not affect only the engine. It potentially requires re-validation or redesign of:

- **High-pressure pumps**
- **Vaporizers / heat exchangers**
- **Valves & actuators**
- **Piping / fittings / flanges**
- **Pressure vessels & accumulators**
- **Instrumentation and pressure transmitters**
- **Sealing systems**
- **Safety / relief systems**

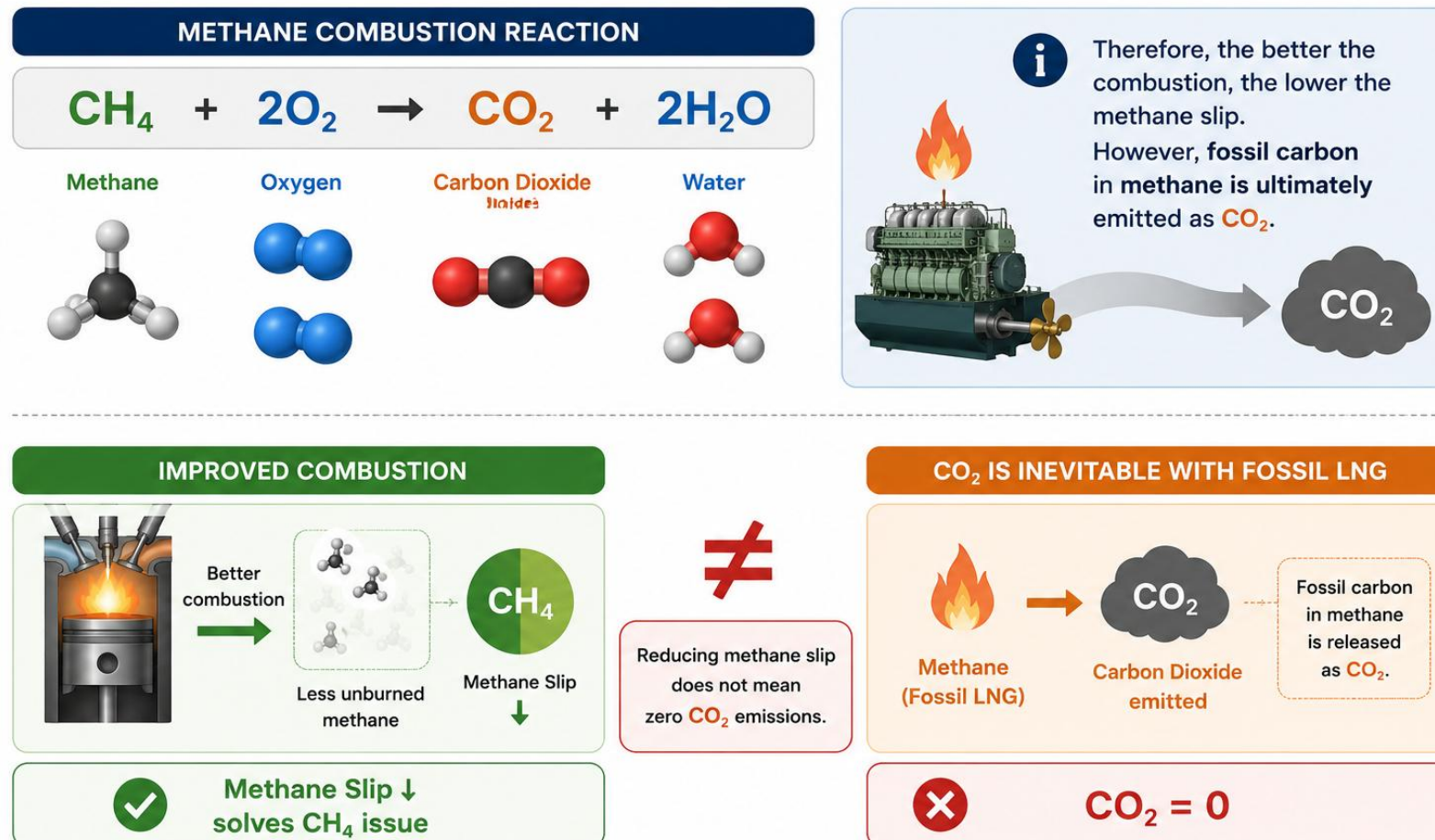
380bar injection is not merely an engine upgrade, it creates a new design requirement across the entire high-pressure LNG fuel supply system.

(Source : Everllence)

5. Key Challenge

Beyond Methane Slip: Tackling LNG's Remaining CO₂

Even with highly efficient LNG engines and negligible methane slip, fossil LNG remains a carbon-containing fuel and therefore cannot achieve net-zero emissions by itself.



Source : ICCT

Beyond Methane Slip: Tackling LNG's Remaining CO₂

Improve Engine Efficiency

Less LNG consumed → Less CO₂ generated

Shift the Fuel Carbon Source

Fossil LNG

→ **Bio-LNG**

→ **e-LNG / synthetic methane**

Capture the Remaining CO₂

LNG Engine

→ Exhaust Gas

→ **Onboard Carbon Capture**

→ Liquid CO₂ Storage / Offloading



Onboard Carbon Capture and Storage System

The next challenge for LNG is no longer only methane slip, it is how to eliminate or manage the remaining fossil CO₂ emissions.

An aerial photograph of a tropical coastline. The left side of the image is filled with a dense, lush green forest. The right side shows a calm, turquoise body of water. A small wooden pier or dock is visible at the bottom left, extending into the water. The entire image is framed by a thin white border.

05

LNG Dual-Fuel Shipbuilding

5. LNG Dual-Fuel Shipbuilding

Korean shipbuilder fleet status (excluding LNGC)

LNG DF feet delivered by shipyard (2020-2025)

	2020	2021	2022	2023	2024	2025	TOTAL
HD HHI Group	4	10	10	14	30	17	85
Hanwha Ocean			4	14	7	13	38
SHI	2	8	4	13	10	1	38
DHS				2	2		4
HJSC						2	2
K-SB					1		1
TOTAL	6	18	18	43	50	33	168

LNG DF feet delivered by ship type (2020-2025)

	2020	2021	2022	2023	2024	2025	TOTAL
Container	2	4	8	23	42	30	109
Tanker	2	12	8	17	4		47
Bulker	2	1	1	2			6
Car Carrier		1	1		1	3	6
TOTAL	6	18	18	43	50	33	168

- Major shipbuilders are building LNG DF vessels.(95.8%)
- Containership is main vessel in Korean shipyard.
- Everllence(138) and WinGD(30) dominate the engine market.

5. LNG Dual-Fuel Shipbuilding

Korean shipbuilder Orderbook (excluding LNGC)

LNG DF feet to be delivered by shipyard (2026-2029)

	2026	2027	2028	2029	TOTAL
HD HHI Group	3	17	35	12	67
Hanwha Ocean		4	15	4	23
SHI			2	7	9
TOTAL	3	21	52	23	99

LNG DF feet to be delivered by ship type (2026-2029)

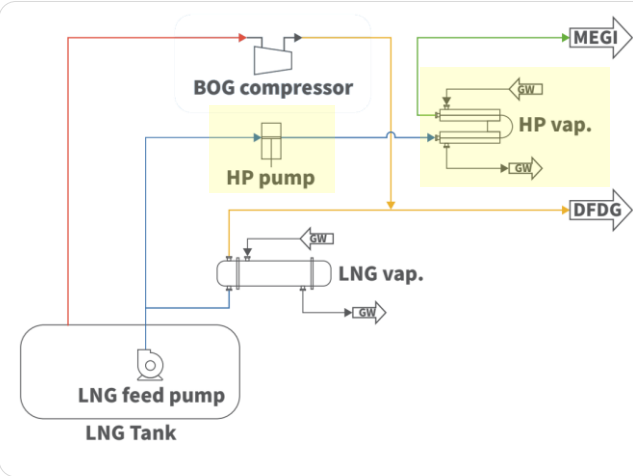
	2026	2027	2028	2029	TOTAL
Container	2	18	47	21	88
Car Carrier	1	3	3		7
LCO2 Carrier			1	1	2
Tanker			1	1	2
TOTAL	3	21	52	23	99

- Only major shipbuilders are building LNG DF vessels.
- Containership is main vessel in Korean shipyard.
- Small and Medium shipbuilders are building conventional fueled ships.

5. LNG Dual-Fuel Shipbuilding

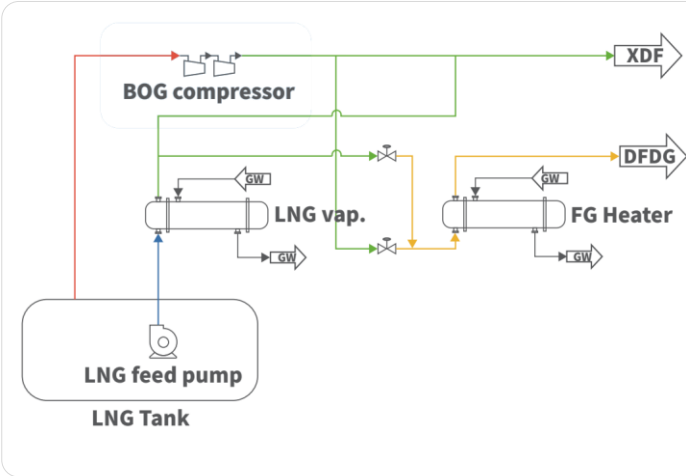
Key equipment for LNG DF vessels and Localization Status

FGSS configurations by Engine Type and Fuel Supply Concept



Everllence

- High-pressure System (~300 bar)
- Requires HP Pump & HP Vaporizer
- High CAPEX / High Efficiency



WINGD

- Lower Pressure Supply (~16 bar)
- LP Feed pump & FG Heater required
- Lower CAPEX / Simpler System

Key Components & Localization Status

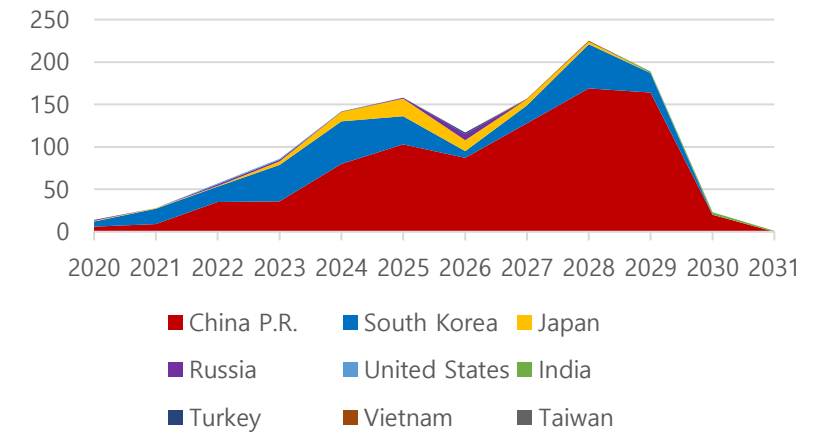
LNG Storage Tank	Mostly Locally Sourced
BOG Compressor	
LNG Vaporizer	
Glycol Water System	
LNG Feed Pump (Low pressure)	Foreign-Dominated, with Emerging Local Suppliers
LNG High Pressure Pump	

5. LNG Dual-Fuel Shipbuilding

Global LNG DF Shipbuilding – The Shift Toward China

LNG DF deliveries by shipbuilding country - 2-stroke, excluding LNGC

	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	TOTAL
China P.R.	6	9	35	36	80	103	87	128	169	164	20		852
South Korea	6	18	18	43	50	33	8	21	52	23			282
Japan	1	1	1	4	11	21	13	7	3				62
Russia	1		2	2		1	7						13
United States			1	1									10
India										2	3	1	6
Turkey							2						6
Vietnam								1	1				2
Taiwan					1								1
TOTAL	14	28	57	86	142	158	117	157	225	189	23	1	1,234



China Leads the Market

- China represents 69% of total LNG DF vessel deliveries, compared with 23% for South Korea.

Further Shift Toward China

- China's share is expected to rise further, reaching 74-87% of annual deliveries during 2026-2029.

5. LNG Dual-Fuel Shipbuilding

Global LNG DF Shipbuilding – The Shift Toward China

LNG DF Vessel Delivery Share (2020-2031) – China and Korea, including LNGC

	China	Korea	China Share	Korea Share
Excl. LNGC	852	282	69.00%	22.90%
Incl. LNGC	998	810	52.20%	42.40%

Korea Share

22.9% → 42.4%

when LNGC are included

	China	Korea	China Share	Korea Share
2026	112	79	52.60%	37.10%
2027	146	85	61.10%	35.60%
2028	191	105	63.20%	34.80%
2029	185	58	75.50%	23.70%

However,

even including LNG carriers—one of Korea's strongest segments—China accounts for more than half of total LNG DF vessel deliveries.

China is becoming the largest addressable market for LNG FGSS and its key components.

Global LNG DF Shipbuilding – The Shift Toward China

Different Procurement Structures

Korea

Individual equipment purchased separately
→ Integration & Packaging by Shipyard

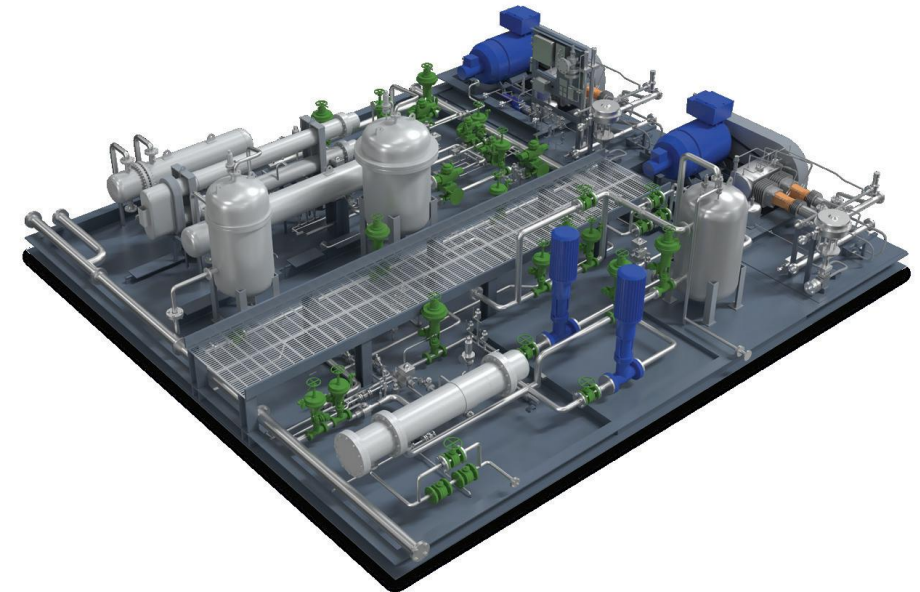
China

Complete FGSS purchased from Package Supplier
→ Integrated FGSS package delivery to Shipyard

Leading FGSS Suppliers in China

SUNRUI / QIYAO / CGT / LGM / HEADWAY, etc.

SUNRUI (CSSC-affiliated) is one of the leading FGSS suppliers in China.



Typical FGSS Package Configuration

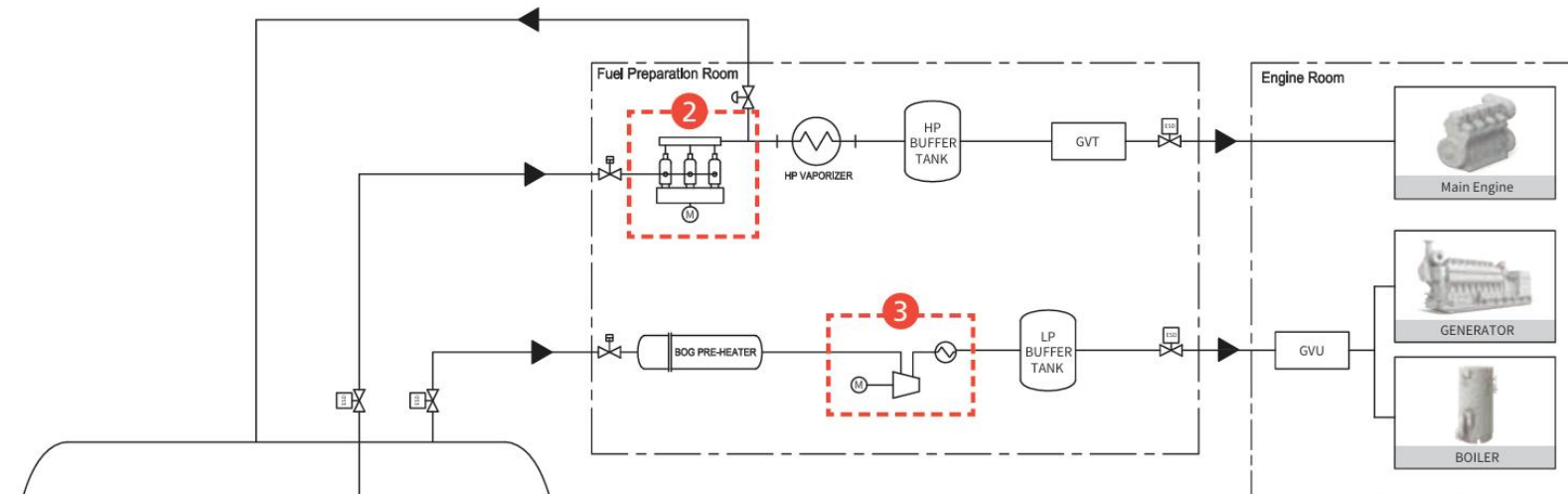


06

PANASIA EM Products for LNG DF ships

6. PANASIA EM Products for LNG DF ships

Product Overview



LNG Fuel Supply Pump(LP)
Initial LNG feed to FGSS system

LNG Fuel Supply Pump(HP)
High pressured LNG for MEGI engine

BOG Compressor
Boil off gas supply to XDF engine



6. PANASIA EM Products for LNG DF ships

LNG Fuel Supply Pump (LP)

Application

- LNG fuel supply to FGSS
- LNG transfer between storage tanks

Specification

Type	Unit	Centrifugal multi-stage submerged
Capacity	m ³ /h	1~36
Total head	mlc	15~450
Liquid temperature	°C	-163
Discharge bore	Mm	-
Power(motor)	kW	5~95
NPSHr	m	0.15~1.5
Differential pressure	bar	20
Design pressure	barg	30
Speed range	rpm	1,200~7,200



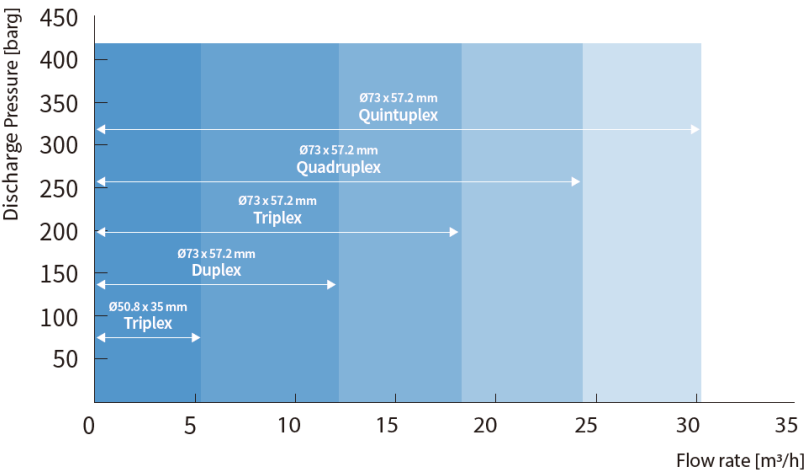
Line-up

Model	Max. Flow Rate [m ³ /hr]	Max. Operating Pressure [barg]
PEH-L20N	24	20
PEH-L30N	36	

6. PANASIA EM Products for LNG DF ships

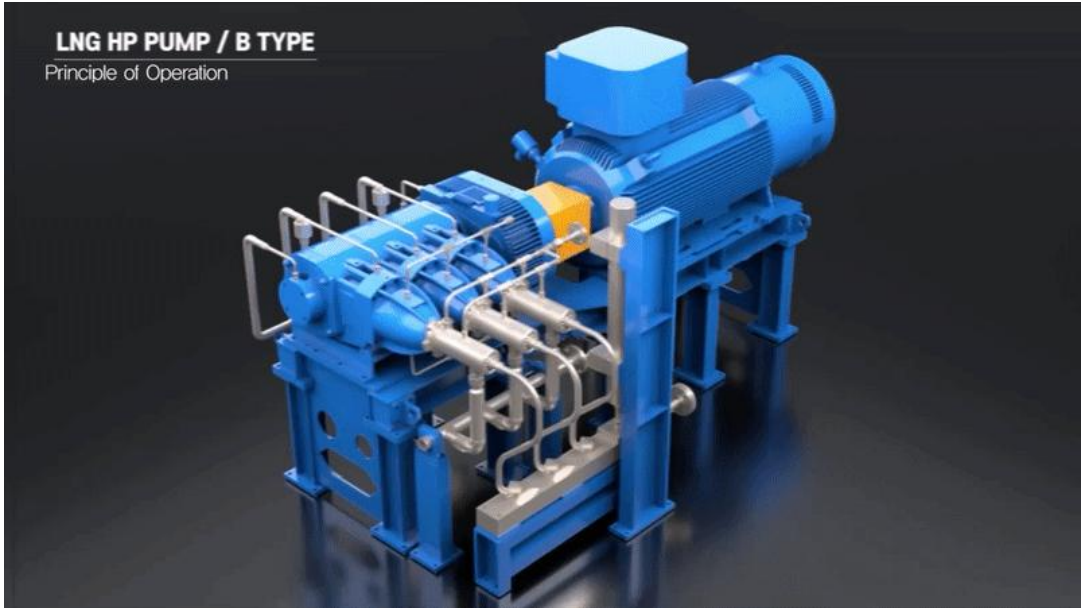
LNG Fuel Supply Pump (HP)

Performance Curve



Specification

Type	Unit	Reciprocating
Capacity	m³/h	1.5~30
Design pressure(Suction)	barg	10
Design pressure (Discharge)	barg	420
Design density	kg/m³	450
Liquid temperature	°C	-163
Power(motor)	kW	35-475



Line-up

Type	Max. Flow Rate [m³/hr]	Max. Operating Pressure [barg]
PEH-L05N	5	325 or 380
PEH-L10N	10	
PEH-L15N	15	
PEH-L20N	20	
PEH-L25N	25	
PEH-L30N	30	

6. PANASIA EM Products for LNG DF ships

BOG Compressor



Specification

Type	Oil injected screw type
Fluid	CH ₄ , N ₂ , CO ₂ , etc.
Capacity	Max. 5500Nm ³ /hr
Suction Pressure	In accordance with customer requirements
Suction Temperature	5~50°C
Discharge Pressure	Max. 16 barg
Discharge Temperature	Max. 100°C

Line-up

Category	Unit	PC13	PC14	PC21	PC22	PC30
Power range	Kw	30~75	37~110	75~200	90~250	132~400
	hp	40~100	50~150	100~270	120~330	180~550
Inlet flow range	m ³ /min	3~8	4~13	7~20	9~22	17~50
	cfm	105~280	140~460	250~700	315~775	600~1760
Male rotor speed range	rpm	1500~6600	1300~6500	1000~5000	900~3900	700~3400
Max allowable working pressure	barg	20	20	20	20	20
	psig	290	290	290	290	290

6. PANASIA EM Products for LNG DF ships

Track Records – LP pump

No.	Date	Project	Capacity	Qty. (Set)	Fluid	Customer	End-User	Class.
1	2015.06	10,000TEU CNTR Grade LNG Demonstration facility	20 m ³ /h	1	LNG	STX Offshore & Shipbuilding (now K Shipbuilding)	STX Offshore & Shipbuilding (now K Shipbuilding)	-
2	2015.11	Incheon City LCNG Refueling Station	20 m ³ /h	1	LNG	Sewon Industry	Sewon Industry	-
3	2016.10	DSME Energy System Test Center	20 m ³ /h	2	LN2 / LNG	Wonil T&I Co., Ltd.	DSME (now Hanwha Ocean)	-
4	2018.09	Mongolia LCNG Refueling Station	20 m ³ /h	6	LNG	Entec I Co., Ltd.	HYOSUNG	-
5	2019.07	DSME Energy System Test Center	20 m ³ /h	1	LN2 / LNG	DSME (now Hanwha Ocean)	DSME (now Hanwha Ocean)	-
6	2022.12	Low-Pressure LNG Pump (25 m ³ /h) for LNG Fuel Supply System	27 m ³ /h	1	LN2 / LNG	DSME (now Hanwha Ocean)	DSME (now Hanwha Ocean)	(TA) KR
7	2022.12	NG Fuel Supply System Technology for Coastal Vessels	5.5 m ³ /h	1	LN2 / LNG	DongHwa Entec	DongHwa Entec	(TA) KR
8	2026.07	Subcooling Re-liquefaction System(SRS) Project	23.2 m ³ /h	1	LNG	Hanwha Ocean Ecotech	Hanwha Ocean	-

6. PANASIA EM Products for LNG DF ships

Track Records – HP pump

No.	Date	Project	Capacity	Qty. (Set)	Fluid	Customer	End-User	Class.
1	2016.05	DSME Energy System Test Center	5.5 m ³ /h	2	LN2 / LNG	Sunbo Industries	DSME (now Hanwha Ocean)	-
2	2017.04	Incheon City LCNG Refueling Station	5.5 m ³ /h	1	LNG	Sewon Industry	Sewon Industry	-
3	2017.06	Pohang City LCNG Refueling Station	5.5 m ³ /h	1	LNG	Korea Gas Corporation	Korea Gas Corporation	-
4	2018.08	Pohang City LCNG Refueling Station	5.5 m ³ /h	2	LNG	Entec I Co., Ltd.	Korea Gas Corporation	-
5	2019.07	DSME Energy System Test Center	16.6 m ³ /h	2	LN2 / LNG	DSME (now Hanwha Ocean)	DSME (now Hanwha Ocean)	-
6	2019.08	Daejeon City LCNG Refueling Station	5.5 m ³ /h	1	LNG	ILSHIN Gas Co., Ltd.	Korea Gas Corporation	-
7	2020.02	Kimpo City LCNG Refueling Station	5.5 m ³ /h	2	LNG	Entec I Co., Ltd.	Sunjin Network Co., Ltd.	-
8	2020.09	Pohang City LCNG Refueling Station	5.5 m ³ /h	2	LNG	Korea Gas Corporation	Korea Gas Corporation	-
9	2020.11	Daejeon LCNG Refueling Station	5.5 m ³ /h	1	LNG	Korea Gas Corporation	Korea Gas Corporation	-
10	2023.10	Kimpo LCNG Refueling Station	5.5 m ³ /h	1	LNG	Sunjin Network Co., Ltd.	Sunjin Network Co., Ltd.	-
11	2023.12	Pohang City LCNG Refueling Station	5.5 m ³ /h	1	LNG	Korea Gas Corporation	Korea Gas Corporation	-
12	2024.04	17K BULK CARRIER - KOMERI MPV	1.6 m ³ /h	1	LNG	VALMAX	KOMERI	(TA) KR
13	2024.05	Incheon City LCNG Refueling Station	5.5 m ³ /h	1	LNG	Sewon Industry	Sewon Industry	-
14	2024.05	Hanwha Ocean Energy System Research Center	2.6 m ³ /h	1	LCO2	Hanwha Ocean Ecotech	Hanwha Ocean	-
15	2025.02	Daejeon LCNG Refueling Station	5.5 m ³ /h	1	LNG	Korea Gas Corporation	Korea Gas Corporation	-
16	2016.11	High-Pressure Pump for Marine LNG Supply System	16.6 m ³ /h	1	LN2 / LNG	DSME (now Hanwha Ocean)	DSME (now Hanwha Ocean)	(TA) KR, DNV
17	2023.06	High-Pressure Pump for Marine LNG Supply System	3.3 m ³ /h	1	LN2 / LNG	DSME (now Hanwha Ocean)	DSME (now Hanwha Ocean)	-
18	2025.12	1-Ton Class LH ₂ Storage Tank and Refueling/Supply Technology	1,100 m ³ /h	1	LH2	Cryos Co., Ltd.	Cryos Co., Ltd.	-

6. PANASIA EM Products for LNG DF ships

Flexible FGSS Integration

① Pump Only

HP LNG Pump Package

→ For FGSS makers with their own skid integration capability

② Pump + Key Components

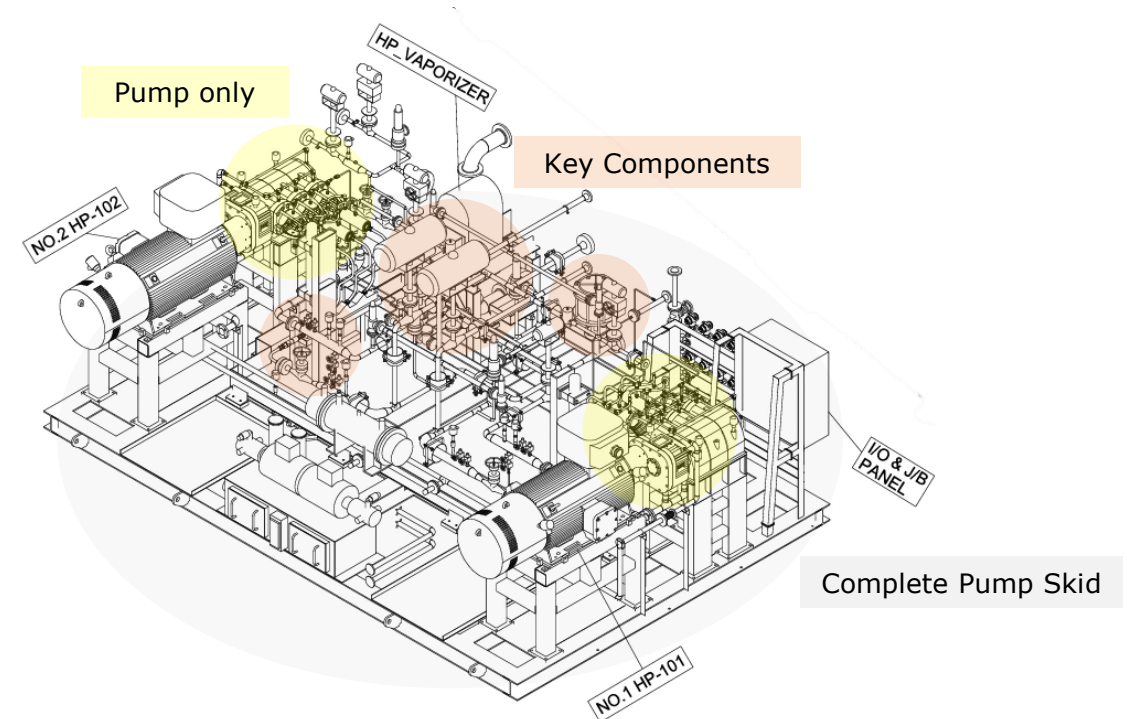
Pump + Piping / Valves / Instruments

→ Customized scope based on FGSS maker requirements

③ Complete Pump Skid

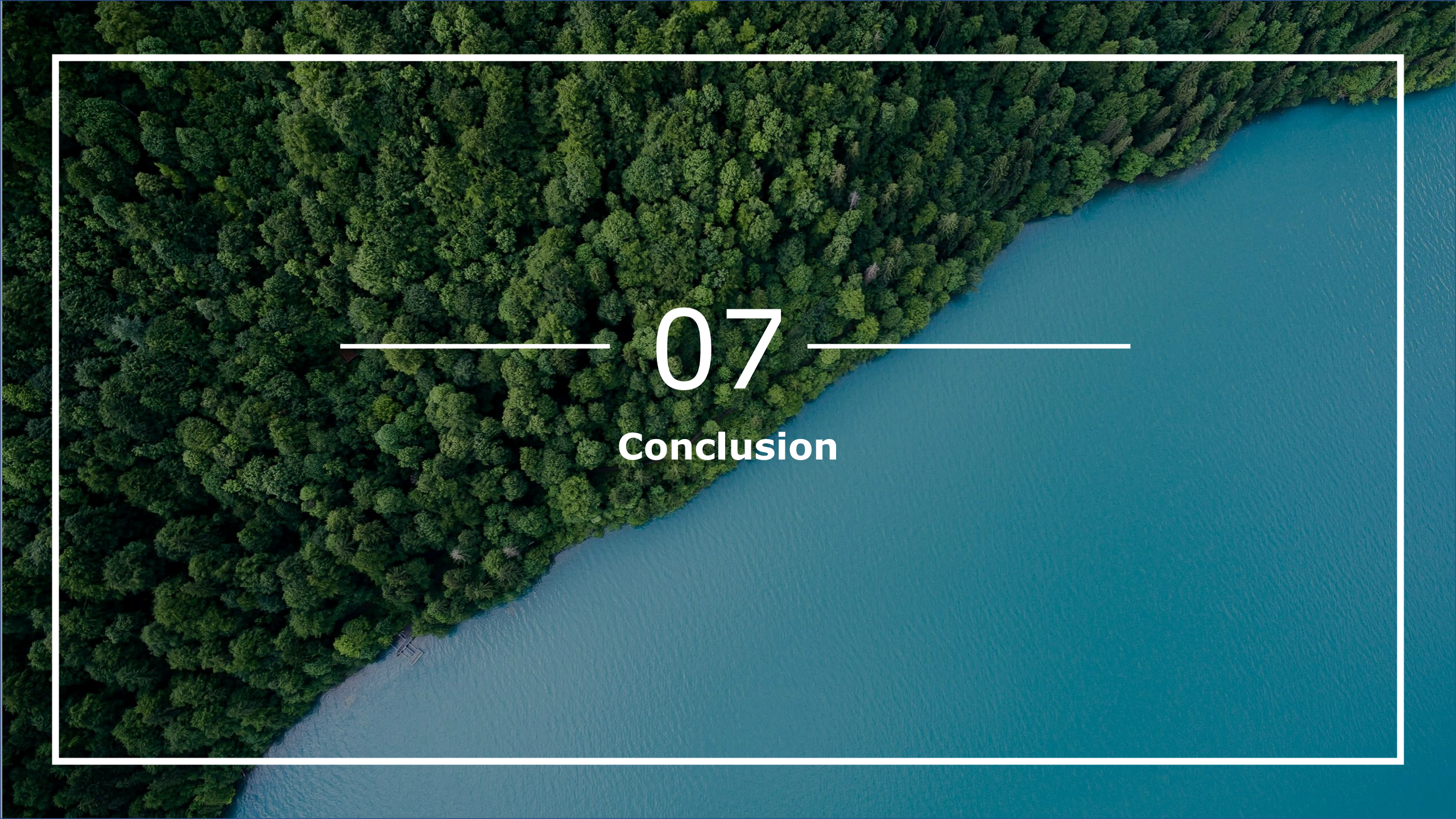
Fully Integrated Pump Skid Package

→ Ready for integration into the FGSS



380 bar Standard Preparation

Model (Capacity)	Progress					
	Design	Procurement	Manufacturing	Installation	Performance Test	Class Approval
HP pump (5m ³ /h)	✓	✓	✓	✓	Planned for Q3 2026	As required
HP pump (15m ³ /h)	✓	✓	✓	✓		

An aerial photograph showing a dense, lush green forest on the left side, which meets a calm, turquoise lake on the right. The forest is composed of many small, rounded tree canopies. The lake's surface is slightly rippled. A small wooden dock or pier is visible at the bottom left, extending into the water. The entire scene is framed by a white border.

07

Conclusion

7. Conclusion

01 MARKET	02 OPPORTUNITY	03 TECHNOLOGY	04 PANASIA EM
LNG Remains a Key Transition Fuel	Global LNG DF Market Continues to Grow	380 bar Creates a New Market Opportunity	Ready for Global FGSS Collaboration
Proven technology & infrastructure	687 vessels in 2027–2029	New standard for ME-GI HP fuel supply	LP / HP Pump + BOG Compressor
Practical pathway toward net zero	China emerging as the key market	Localization opportunity for LNG pumps	Flexible supply & integration concept

PANASIA EM will become a reliable global partner in LNG fuel supply solutions through proven technology, flexible integration and close collaboration with FGSS makers.



Thank you
for your attention

Catalog
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